

Smooth Nonlinear ESO-Based Model-Free Predictive Current Control With an Extended Control Set for SPMSM Drives

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Abstract—This article proposes an improved nonlinear extended state observer-based model-free predictive current control (MFPCC) with an extended control set. First, the non-smoothness problem of conventional nonlinear extended state observer model-free solutions is explained. By constructing a high-smoothness polynomial series at the transition points of the piecewise nonlinear function, an improved nonlinear extended state observer is designed to enhance the performance of the MFPCC, and relevant stability proofs are derived by the Routh criterion. Furthermore, a newly designed extended control set based on the voltage vectors (VVs) at the center of triangular subregions (TSs) of the modulation area is proposed to improve the output voltage accuracy. Based on the geometrical characteristics of the subregion center VVs, a fast-layering selection method is proposed to reduce the complexity of determining the optimal VV. Finally, experimental results are presented to verify the proposed method's feasibility and effectiveness.

Index Terms—Extended control set, nonlinear extended state observer, predictive current control, ultra-local model

I. INTRODUCTION

PERMANENT magnet synchronous motors (PMSMs) are widely used in electric vehicles, underwater propellers, and other fields due to their high-power density, energy saving, and high performance [1], [2]. In recent years, model predictive control (MPC) based on a system model has become an increasingly promising approach [3], [4], [5].

However, control performance of the MPC methods rely on the physical parameters of the model, which are often subject to time-varying influences by external conditions [6], [7]. Among the many MPC methods for PMSM-driven systems,

model-free predictive current control (MFPCC) has recently attracted a lot of attention as a method to reduce the sensitivity to model parameters, resulting in a higher robustness of MPC [8]. In PMSM drives, MFPCC is usually classified into two categories: the current error update-based MFPCC [9], [10] and the ultra-local model-based MFPCC [11]. Lin et al. [9] initially proposed an MFPCC method that completes current prediction by storing and updating a lookup table of current errors under different voltage vectors (VVs). Although this method is conceptually simple, the performance of current control may deteriorate due to stagnation in the update of current errors, and the stability of MFPCC cannot be guaranteed. To address this issue, Ma et al. [10] proposed an update mechanism based on the current gradients over two consecutive periods. In this mode, the current errors for all fundamental vectors can be obtained simultaneously at each moment. Another MFPCC method, based on the ultra-local model theory proposed in [12], simplifies the current model of PMSM as a first-order model of a single-input, single-output system and substitutes partial state variables and disturbances. In [13], an algebraic identification method is combined with the ultra-local model theory of PMSM. However, this method suffers from a significant computational burden. To achieve a better fault-tolerant control performance, the ultra-local model of PMSM is estimated by a nonsingular terminal sliding mode controller (SMC) in [14], however, the chatter problem of the SMC deteriorates the estimation performance. In [15], a technique utilizing linear ESO (LESO) to estimate the unknown part of the ultra-local model of PMSM is proposed, which enhances robustness and effectively simplifies the estimation process. Compared to LESO, nonlinear ESO (NLESO) demonstrates more significant potential in coping with uncertain disturbances and improving dynamic performance. This is attributed to the adaptive nature of the nonlinear function within NLESO, which adjusts based on specified error bounds, thus offering higher estimation accuracy [16]. Zhang et al. [17] proposes an NLESO-based MFPCC to improve the robustness of MPCC. However, the nonlinear function of NLESO is not smooth at the boundaries of the piecewise function, i.e., the higher order derivatives are discontinuous at the boundaries of the segmented function, which can lead to nonsmoothness problems in the estimation of the ultra-local model.

Moreover, the limitation of the output VV of the voltage-based two-level inverter leads to the MPC's finite control set, which results in its poor steady-state performance problem.

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Because bigger quantities of VVs are available, more complex power electronic topologies have been adopted [18], [19]. However, these topologies require additional hardware devices, involve dc-link capacitor-voltage balancing issues, and require more complex algorithms. Multivector methods [20], including duty cycle control [21], are also used to improve steady-state performance. Although good steady-state performance is achieved, those methods require complex implementation steps, resulting in a significant computational burden. Zhou et al. [22] introduce an extended control set with additional virtual VVs for duty cycle calculations. While this approach achieves good steady-state performance, it requires three complex execution steps that impose a significant computational burden. Recently, the discrete space vector modulation (DSVM) method has been recognized as an attractive approach for utilizing the discrete architecture adaptability of the inverter to extending the control set [23], [24], [25]. In [23], the DSVM technique was first applied to the DTC method, which extends the number of feasible VVs by subdividing a control period into equal intervals which is easy implemented in control systems. This way, steady-state performance is improved, and current ripples are reduced. Although the candidate VV set generated by DSVM is expanded, resulting in increased precision of output vectors, it also increases the selection difficulty. For example, when $N_X = 3$, the number of candidate VVs is increased from 8 to 38. Determining the optimal VV within a control period becomes challenging, especially in DSP control systems [24]. To address this problem, [25] reduces the number of 37 candidate VVs to 12 by evaluating six specific VVs in the first stage. This effectively reduces the number of alternative VVs but consumes additional execution time due to continuously updating the current gradients corresponding to the VVs in each control period. In [26], the reference VV (RVV) is first determined according to the deadbeat direct torque flux control principle (DB-DTFC). The distance on the three auxiliary lines determines the optimal triangular subregion (TS). Finally, the optimal VV on the vertexes of the optimal TS is determined by the cost function. Similarly, the deadbeat control principle is employed in [27] to obtain RVVs and reduce the 38 candidate VVs to 2, 4, or 6. Although methods mentioned above have made improvements in improving the steady-state performance and reducing computational burden, it is necessary to consider more accurate voltage output and the problem of the parameter sensitivity.

In this article, an improved NLESO-based MFPC with a newly designed-extended control set is proposed to improve steady-state performance and enhance robustness of MPCC. The main contributions of this article include below.

- 1) A high-smoothness NLESO-based MFPC is proposed to solve the nonsmoothness problem of nonlinear function in the conventional NLESO-based MFPC. The proposed NLESO fits a new function based on polynomial series to improve the problem of discontinuity of the derivatives of each order in the original nonlinear function and improve the smoothness.
- 2) Motivated by the DSVM theory [23], [26], taking advantage of the excellent geometrical properties of homo-

geneously divided positive TSs, the subregion-centered VVs (SCVVs) are defined as elements of a newly designed extended control set. This extended control set can increase the VV number of the DSVM-based extended control set from 38 to 62, thus enhancing the output voltage accuracy and improving the steady-state control performance.

- 3) Meanwhile, to simplify the selection process and reduce the computational burden, based on SCVV's geometrical characteristics, this article proposes an optimal VV selection method. The availability of the optimal basic VV that satisfies the conditions is first determined based on the relevant distance. Otherwise, the optimal virtual VV's corresponding duty cycle will be directly determined based on the projection of the RVV to the three auxiliary lines without any cost function traversal. In addition, the switching sequence of the virtual VV is optimized to reduce the switching frequency.

II. MATHEMATICAL MODEL OF SPMSM AND THE CONVENTIONAL NLESO-BASED ULTRA-LOCAL MODEL

A. Mathematical Model of SPMSM

The mathematical model of SPMSM in the dq -axis synchronous rotating reference frame can be expressed as follows:

$$\begin{cases} u_d = L_s \frac{di_d}{dt} + R_s i_d - \omega_e L_s i_q \\ u_q = L_s \frac{di_q}{dt} + R_s i_q + \omega_e (L_s i_d + \psi_f) \end{cases} \quad (1)$$

where R_s and L_s represent the stator resistance and the stator inductance; i_d and i_q are the dq -axis currents; ω_e is the electrical rotor speed (in rad/s); ψ_f is the flux linkage. And u_d and u_q represent the dq -axis voltages.

In this article, the SPMSM is driven by a three-phase two-level inverter, and the basic VVs of a three-phase two-level inverter and can be described as

$$V_i = \frac{2}{3} U_{dc} \left(S_a + S_b e^{\frac{2}{3}\pi i} + S_c e^{-\frac{2}{3}\pi i} \right) \quad (2)$$

where V_i (i is an integer between 0 and 7) is the expectation of output VVs. S_a , S_b , and S_c represent the switching function of the three-phase two-level voltage-source inverter, respectively.

B. Conventional NLESO-Based Ultra-Local Model and its Limitation

As discussed in [6], inaccurate motor parameters can significantly impact the control effectiveness of MPCC. To address the drawbacks associated with parameter dependency, a model-free approach has been employed during the RVV information acquisition.

A first-order single-input single-output ultra-local model can be represented as [12]

$$\dot{y} = \alpha u + F \quad (3)$$

where u and y represent the input and output of the first-order system, α is the coefficient of the scale factor, and F is the part that contains uncertainty.

Then, based on the model-free control principle proposed in [13], the ultra-local model of PMSM can be described as

$$\frac{d}{dt}i_{d,q} = F_{d,q} + \alpha u_{d,q} \quad (4)$$

where $F_{d,q}$ contains uncertainties caused by parameter changes and external disturbances in the d -axis and q -axis, and α can be set to $1/L_s$ as the initial value.

Based on the aforementioned ultra-local model, taking an arbitrary model of the dq -axis as an example and employ the following conventional NLESO to estimate F :

$$\begin{cases} e_1 = z_1 - x_1 \\ \dot{z}_1 = z_2 + \alpha u_s - \beta_1 \text{fal}(e_1, \varepsilon, \delta) \\ \dot{z}_2 = -\beta_2 \text{fal}(e_1, \varepsilon, \delta) \end{cases} \quad (5)$$

where x_1 is the actual current in d -axis or q -axis, z_1 is the estimation of the x_1 , e_1 is the estimate error of x_1 , β_1, β_2 are the gains of the ESO and the nonlinear function $\text{fal}(e_1, \varepsilon, \delta)$ is defined as follows [16]:

$$\text{fal}(e_1, \varepsilon, \delta) = \begin{cases} \zeta(e_1, \varepsilon, \delta) = \frac{e_1}{\delta^{(1-\varepsilon)}}, |e_1| \leq \delta \\ \eta(e_1, \varepsilon, \delta) = |e_1|^\varepsilon \text{sign}(e_1), |e_1| > \delta \end{cases} \quad (6)$$

where ε is a nonlinear factor and δ characterizes the size of the nonlinear interval. The two parameters determine the shape of the nonlinear function. The value range of ε is a constant between 0 and 1 and δ is a filtering parameter. The $\text{sign}(\cdot)$ denotes a sign function.

From the conventional NLESO (5) and (6), the first-order derivative function of the nonlinear fal function when $\varepsilon > \delta$, i.e., when e_1 tends to δ^+ , is

$$\left. \frac{d}{de_1} \text{fal}(e_1, \varepsilon, \delta) \right|_{e_1 \rightarrow \delta^+} = \varepsilon \delta^{\varepsilon-1} \quad (7)$$

where δ^+ is the approaching δ from the right side. Similarly, when $0 < e_1 < \delta$, i.e., when e_1 tends to δ^- , the first-order derivative function of the nonlinear fal function is

$$\left. \frac{d}{de_1} \text{fal}(e_1, \varepsilon, \delta) \right|_{e_1 \rightarrow \delta^-} = \frac{1}{\delta^{1-\varepsilon}} \quad (8)$$

where δ^- is the approaching δ from the left side.

From the above two equations, at $e_1 = \delta^+$, the left and right derivatives of the fal function are unequal, i.e., the first-order derivative of the fal function is discontinuous. Therefore, the fal function is not smooth at $e_1 = \delta$, and similar situation will happen around $e_1 = \delta^-$.

From the above two equations, the dynamic adjustment of z_2 can be approximated as follows:

$$\Delta z_2 = -T_s \beta_2 \left(\frac{1}{\delta^{1-\varepsilon}} - \varepsilon \delta^{\varepsilon-1} \right). \quad (9)$$

When setting the observer parameters in the control system, the bandwidth β_2 is often large enough to get better bandwidth [29]. Thus, the abrupt changes in the feedback errors are induced as e_1 approaches the neighborhood of $\pm\delta$, which affects the smoothness of the dynamic regulation of the z_2 observation process. Especially in rapidly changing systems, such abrupt changes can lead to lags and oscillations in the adjustment of z_2 , thus affecting the system's dynamic

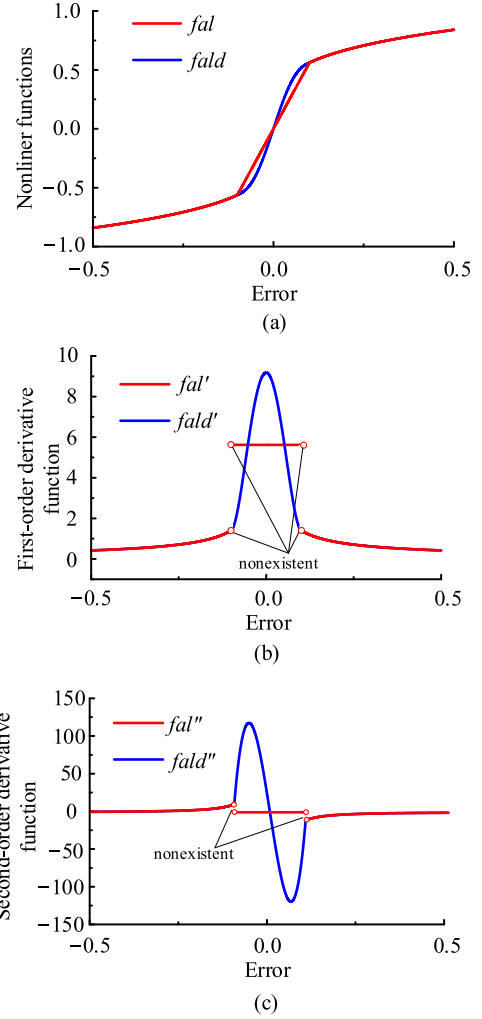


Fig. 1. Comparison of the $\text{fal}(e_1, \varepsilon, \delta)$ and the $\text{fald}(e_1, \varepsilon, \delta)$. (a) Two functions, (b) first-order derivative functions, and (c) second-order derivative functions.

performance. Due to the dependence of the observation performance, the effect of poor estimation of z_2 can weaken the performance of the model-free control system. Therefore, reducing the nonsmoothness of the nonlinear function at transition points in NLESO is necessary.

III. HIGH SMOOTHNESS NLESO-BASED MODEL-FREE RVV ACQUISITION

A. New Designed NLESO and RVV Acquisition

This section designs a new function to replace the nonlinear function in NLESO, improving F 's estimation performance in the ultra-local model and solving traditional NLESO's nonsmoothness at the error boundary.

Expand the $\xi(e_1, \varepsilon, \delta)$ function in the fal function into a convergent polynomial series $\kappa(e_1, \varepsilon, \delta)$ composed of polynomials

$$\kappa(e_1, \varepsilon, \delta) = \sum_{i=1}^n a_i e_1^i \quad (10)$$

where a_i is the i th coefficient.

According to the strict definition of a smooth function, a necessary and sufficient condition for smoothness is that all derivatives of the function exist and are continuous. To ensure

the overall smoothness of the fal function, the designed $\kappa(e_1, \varepsilon, \delta)$ function needs to be smooth at the boundary junction with the $\eta(e_1, \varepsilon, \delta)$ function, that is, the designed function should be equal to the derivative function of the $\eta(e_1, \varepsilon, \delta)$ function

$$\eta^n(e_1, \varepsilon, \delta) = \kappa^n(e_1, \varepsilon, \delta) \quad (11)$$

where the superscript n represents the n -order derivative of the function.

However, too many differentiable expressions in series require higher order polynomials, which means more parameter design is needed and increases the difficulty of observer design. To comprehensively consider the smoothness at the error boundary of the observer's nonlinear function and the design difficulty of actual implementation, this function is designed to a second derivative. Since the second-order derivative function has six boundary conditions, the first six terms of the above infinite series should be taken, that is, the polynomial series $\kappa(e_1, \varepsilon, \delta)$ can be roughly expressed as

$$\kappa(e_1, \varepsilon, \delta) = a_1 e_1 + a_2 e_1^2 + a_3 e_1^3 + a_4 e_1^4 + a_5 e_1^5 + a_6 e_1^6. \quad (12)$$

Since the κ and the η are assumed to be smooth and continuous at the intersection, this implies that the condition of (11) holds at $e_1 = \delta$, then the conditions can be obtained as follows:

$$\begin{cases} \kappa(\delta, \varepsilon, \delta) = \eta(\delta, \varepsilon, \delta) \\ \kappa'(\delta, \varepsilon, \delta) = \eta'(\delta, \varepsilon, \delta) \\ \kappa''(\delta, \varepsilon, \delta) = \eta''(\delta, \varepsilon, \delta). \end{cases} \quad (13)$$

Due to the odd function nature of the κ , the coefficients of the even terms in the κ are all zero. Bringing the derivatives of each order of the κ and the η into the above equation, the following conditions hold:

$$\begin{cases} a_1 \delta + a_3 \delta^3 + a_5 \delta^5 = \delta^\varepsilon \\ a_1 + 3a_3 \delta^2 + 5a_5 \delta^4 = \varepsilon \delta^{\varepsilon-1} \\ 6a_3 \delta + 20a_5 \delta^3 = \varepsilon(\varepsilon - 1) \delta^{\varepsilon-2}. \end{cases} \quad (14)$$

The above equation is a system of ternary equations with coefficients a_1, a_3 , and a_5 about constants e_1 and δ . By solving the above system of equations, it is obtained that the coefficients a_1, a_3 , and a_5 are

$$\begin{cases} a_2 = a_4 = a_6 = 0 \\ a_1 = \frac{(\varepsilon^2 - 8\varepsilon + 15)\delta^{\varepsilon-1}}{8} \\ a_3 = -\frac{(\varepsilon^2 - 6\varepsilon + 5)\delta^{\varepsilon-3}}{4} \\ a_5 = \frac{(\varepsilon^2 - 4\varepsilon + 3)\delta^{\varepsilon-5}}{8}. \end{cases} \quad (15)$$

Then, the nonlinear function can be obtained as

$$\text{fald}(e_1, \varepsilon, \delta) = \begin{cases} \kappa(e_1, \varepsilon, \delta) = a_1 e_1 + a_3 e_1^3 + a_5 e_1^5, & |e_1| \leq \delta \\ \eta(e_1, \varepsilon, \delta) = |e_1|^\varepsilon \text{sign}(e_1), & |e_1| > \delta. \end{cases} \quad (16)$$

In Fig. 1, the parameters of conventional NLESO and the proposed one are set as $\varepsilon = 0.25$ and $\delta = 0.1$, and the figures of the first-order derivative function and the second-order derivative function of the $\text{fald}(e_1, \varepsilon, \delta)$ and $\text{fal}(e_1, \varepsilon, \delta)$. The

first-order and the second-order derivative $\text{fal}(e_1, \varepsilon, \delta)$ of conventional NLESO do not exist at the error junction. However, the proposed $\text{fald}(e_1, \varepsilon, \delta)$ does exist which means the proposed $\text{fald}(e_1, \varepsilon, \delta)$ has better smoothness, and the nonsmoothness of F estimation can be alleviated.

Then, the proposed NLESO can be designed as

$$\begin{cases} e_1 = z_1 - x_1 \\ \dot{z}_1 = z_2 + \alpha u_s - \beta_1 \text{fald}(e_1, \varepsilon, \delta) \\ \dot{z}_2 = -\beta_2 \text{fald}(e_1, \varepsilon, \delta). \end{cases} \quad (17)$$

From the ultra-local model (4) and the estimated F , the RVV can be deduced as

$$\begin{cases} u_d^* = \frac{i_d^* - i_d^k}{\alpha T_s} - \frac{1}{\alpha} \hat{F}_d \\ u_q^* = \frac{i_q^* - i_q^k}{\alpha T_s} - \frac{1}{\alpha} \hat{F}_q \end{cases} \quad (18)$$

where u_d^* and u_q^* are the dq -axis components of RVV according to the current deadbeat control principle. The RVV can be transformed into $\alpha\beta$ -axis by the following equation:

$$\begin{cases} u_\alpha^* = u_d^* \cos \theta - u_q^* \sin \theta \\ u_\beta^* = u_d^* \sin \theta + u_q^* \cos \theta \end{cases} \quad (19)$$

where θ is the rotor angle, u_α^* and u_β^* are the $\alpha\beta$ -axis components of RVV. Then, the RVV information of is given by

$$\theta^* = \arctan\left(\frac{u_\beta^*}{u_\alpha^*}\right) \quad (20)$$

$$|u_s^*| = \sqrt{(u_\alpha^*)^2 + (u_\beta^*)^2} \quad (21)$$

where θ^* and $|u_s^*|$ are the reference angle and the amplitude of the RVV.

B. Stability Proof and Parameter Configuration of the Proposed NLESO

Since the constant coefficient of the nonlinear function is fixed, it is assumed to be known here. So, it may be set

$$\text{fald}(e_1, \varepsilon, \delta) = \frac{\text{fald}(e_1, \varepsilon, \delta)}{e_1} e_1 = \chi(e_1) e_1 \quad (22)$$

$$\chi(e_1) = \begin{cases} A_1 = a_1 + a_3 e_1^2 + a_5 e_1^4, & |e_1| \leq \delta \\ A_2 = |e_1|^{\varepsilon-1}, & |e_1| > \delta. \end{cases} \quad (23)$$

Then, the nonlinear observer can be expressed as

$$\begin{cases} e_1 = z_1 - x_1 \\ \dot{z}_1 = z_2 + \alpha u_s - \beta_1 \chi(e_1) e_1 \\ \dot{z}_2 = -\beta_2 \chi(e_1) e_1. \end{cases} \quad (24)$$

Then, the observer equation can be considered as a LESO with variable coefficients. The frequency domain analysis method of linear systems can also be used to analyze the stability of the above observer [16]. According to the observer error, the observer transfer function can be obtained as

$$\begin{aligned} z_1 &= \frac{\beta_1 \chi(e_1) s + \beta_2 \chi(e_1)}{s^2 + \beta_1 \chi(e_1) s + \beta_2 \chi(e_1)} i_s \\ &\quad + \frac{\alpha s}{s^2 + \beta_1 \chi(e_1) s + \beta_2 \chi(e_1)} u_s \end{aligned} \quad (25)$$

$$z_2 = \frac{\beta_2 \chi(e_1) s}{s^2 + \beta_1 \chi(e_1) s + \beta_2 \chi(e_1)} i_s + \frac{\alpha \beta_2 \chi(e_1)}{s^2 + \beta_1 \chi(e_1) s + \beta_2 \chi(e_1)} u_s. \quad (26)$$

The coefficients in the observer are known constants, and the following limit equation holds at the junction of the error function:

$$\lim_{e_1 \rightarrow \pm\delta} A_1(e_1) = \delta^{\alpha-1} = \lim_{e_1 \rightarrow \pm\delta} A_2(e_1). \quad (27)$$

According to the definition of continuity, the function $\chi(e_1)$ is continuous. Due to the definition of continuous functions, the above function must be bounded. According to the conclusion in [28], the observer's transfer function can be considered as a variable coefficient linear system with two finite parameter perturbations, and its stability can be analyzed using the Rols criterion method.

From the transfer function, we know that the characteristic equation of the system is

$$s^2 + \beta_1 \chi(e_1) s + \beta_2 \chi(e_1) = 0. \quad (28)$$

According to the Routh criterion, the stability condition of the system is

$$\begin{cases} \beta_1 \chi(e_1) > 0 \\ \beta_2 \chi(e_1) > 0. \end{cases} \quad (29)$$

There are two situations here. When $e_1 > 0$, the derivative of A_1 can be expressed as

$$A_1' = -2 \frac{(\varepsilon^2 - 6\varepsilon + 5)\delta^{\varepsilon-3}}{4} e_1 + 4 \frac{(\varepsilon^2 - 4\varepsilon + 3)\delta^{\varepsilon-5}}{8} e_1^3. \quad (30)$$

The above derivative function can be rewritten as

$$A_1' = e_1 \left(4 \frac{(\varepsilon^2 - 4\varepsilon + 3)\delta^{\varepsilon-5}}{8} e_1^2 - 2 \frac{(\varepsilon^2 - 6\varepsilon + 5)\delta^{\varepsilon-3}}{4} \right). \quad (31)$$

Since $\delta > e_1 > 0$ holds, the following inequalities always hold:

$$e_1^2 < \delta^2. \quad (32)$$

From (27) and (28), the following inequalities hold:

$$A_1' = e_1 \left(\frac{(\varepsilon^2 - 4\varepsilon + 3)\delta^{\varepsilon-5}}{2} e_1^2 - \frac{(\varepsilon^2 - 6\varepsilon + 5)\delta^{\varepsilon-3}}{2} \right) \leq e_1(\varepsilon - 1)(\delta^{\varepsilon-3}). \quad (33)$$

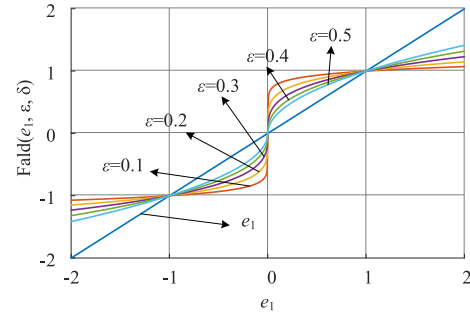
And since the observed error e_1 belongs between 0 and 1, then

$$A_1' \leq 0. \quad (34)$$

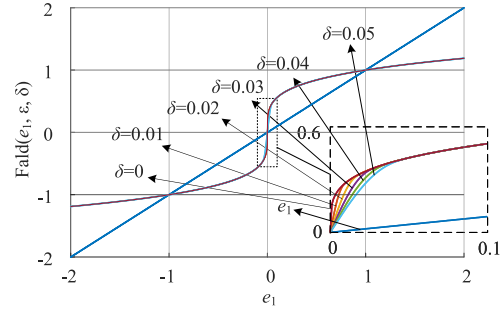
At this time, A_1 takes the minimum value at $e_1 = \delta$, and then the following inequalities hold:

$$A_1(\delta) = \delta^{\varepsilon-1} > 0. \quad (35)$$

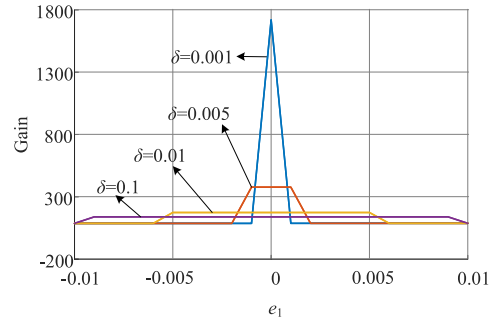
Because the parameters ε and δ are always bigger than zero, A_1 is always positive when error e_1 is between 0 and δ . Similarly, since the A_1 function is an even function, A_1 is



(a)



(b)



(c)

Fig. 2. Relationship between error input and $Fald(e_1, \varepsilon, \delta)$ output under different parameters (a) $\delta = 0.01$ with different ε , (b) $\varepsilon = 0.5$ with different δ , and (c) error gain under different δ .

also positive when e_1 belongs between $-\delta$ and 0, and A_2 is also positive, then $\chi(e_1)$ is a function that is always greater than zero. According to the conditions in the Routh criterion, since $\chi(e_1) < 0$ is always true, the condition for the stability of the newly designed NLESO is that the parameters β_1 and β_2 are strictly greater than zero. Then, the proof is obtained.

As shown in Fig. 2(a), in the proposed smooth NLESO, ε can represent the nonlinearity of the nonlinear function, which is usually set to less than 1. When the error e_1 is less than 1, decreasing ε can increase the nonlinearity of the observation system and thus expand the observation performance under the small error range of e_1 . However, under the minimum error, the excessive gain will affect the system's stability and produce the problem of chattering. So, it is reasonable to set a suitable ε to prevent the nonlinearity from being too strong, leading to high-frequency chattering due to too much gain at a small error [28].

The significance of δ is to limit the high-frequency chatters that occur in the observation due to excessive gains in the

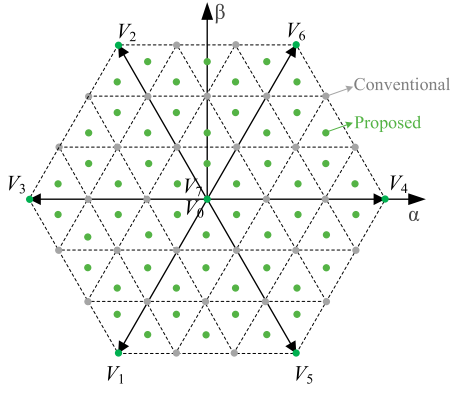


Fig. 3. Extended VV control set in conventional DSVM and the proposed extended VV control set.

error near zero [29]. As shown in Fig. 2(b), with the addition of the δ coefficient term, the gain is smaller in the small error range, thus reducing the high-frequency chatters [30]. Here, considering that the a_3 and a_5 terms are much smaller than a_1 in the usual parameter settings, to facilitate analysis, the term a_1 is used as the dominant term of the fald to show the gain of the fald at different δ and the results are shown in Fig. 2(c). From Fig. 2(c), a small δ can make the gain of the fald function too large and lead to additional chatter near small error ranges. As mentioned in [17], δ can be set to more than ten times the sampling frequency.

Based on the above analysis, for the setting of ε , a comparative conservative value, 0.5, is used in this article to compromise the stability of the system and the nonlinear observation performance. In addition, for the δ setting, $\delta = 0.01$ is used to obtain better observational performance. Regarding β_1 and β_2 , observer bandwidth and noise effects need to be compromised according to the parameter setting method mentioned in [17].

IV. NEWLY DESIGNED EXTENDED CONTROL SET FOR SPMSM DRIVES

A. Newly Designed-Extended Control Set and Geometric Optimal VV Selection Method

Motivated by the DSVM theory, which generates 38 VVs in the extended control set as gray VVs is shown in Fig. 3. To further improve the accuracy of the output voltage, this article proposes a newly designed extended control set, as green VVs shown in Fig. 3, which consists of the SCVVs of each TS of the DSVM-based and the basic VVs. In this way, the number of VVs in the extended control set can be increased from 38 to 62 to further improve the accuracy of the output voltage which can improve the steady-state performance of MFPC.

However, evaluating all candidate VVs within a short control period imposes a significant computational burden especially in microprocessor real-time applications. Therefore, based on the superior geometrical properties of the positive TSs and their center VVs, a geometric perspective-based optimal VV selection method is proposed to simplifying algorithm implementation complexity and reducing the computational burden.

TABLE I
ACTIVE VVs IN EACH SECTOR

θ^*	Sector	V_{avv1}	V_{avv2}
$0 \leq \theta^* < \pi/3$	Sector I	V_4	V_6
$\pi/3 \leq \theta^* < 2\pi/3$	Sector II	V_6	V_2
$2\pi/3 \leq \theta^* < \pi$	Sector III	V_2	V_3
$\pi \leq \theta^* < 4\pi/3$	Sector IV	V_3	V_1
$4\pi/3 \leq \theta^* < 5\pi/3$	Sector V	V_1	V_5
$5\pi/3 \leq \theta^* < 2\pi$	Sector VI	V_5	V_4

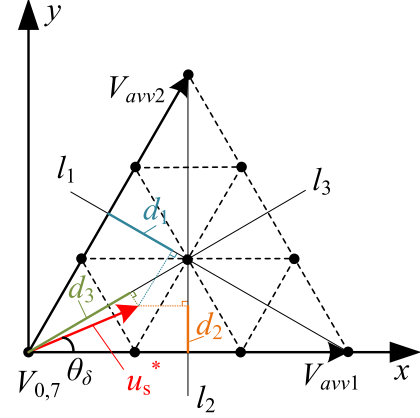


Fig. 4. Layer number determination method.

To narrow the range of candidate VVs, the VV control set within a particular sector is first identified through the reference voltage information. According to Table I, the two active VVs can be determined by the reference angle, θ^* . V_{avv1} is the first active VV in a counterclockwise direction within each sector, while V_{avv2} is the second. To determine the active VVs within a specific sector, it is only necessary to determine the set of candidate VVs that are within the sector.

Due to the superior geometric nature of the SCVVs, the selection of the optimal output VV can be transformed into a problem of selecting the optimal TS. To determine the optimal output VV directly, a VV selection method based on geometric auxiliary lines is proposed. As shown in Fig. 4, take the zero vector as the origin of coordinates, the direction of V_{avv1} as the x-axis, 90° ahead of the direction of V_{avv2} as the y-axis to make a Cartesian coordinate system. Since the rotation of the coordinate axes does not change the amplitude of the RVV, the residual angle θ_δ of at the sector where the RVV resides can be obtained as

$$\theta_\delta = \text{Mod}\left(\frac{\theta_{\text{ref}}}{\pi/3}\right) \quad (36)$$

where the Mod is the remainder function.

Then, make three auxiliary lines, l_1 , l_2 , and l_3 , from the line where the three angle bisectors of the sector of this square triangle lie. As shown in Fig. 4, the distances from RVV to l_1 , l_2 , and l_3 are d_1 , d_2 , and d_3 , respectively, which can be used to determine the specific TS where the optimal VV located in and measure the weight of $V_{\text{opt}1}$, $V_{\text{opt}2}$, and $V_{\text{opt}0,7}$. According to the formula for the distance from a point to a straight line,

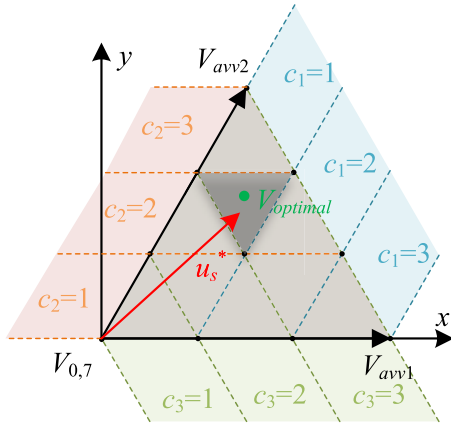


Fig. 5. Optimal TS determination by the layers.

their specific expressions are as follows:

$$\begin{cases} d_1 = |u_s^*| \cos\left(\frac{\pi}{6} + \theta_\delta\right) \\ d_2 = |u_s^*| \sin(\theta_\delta) \\ d_3 = |u_s^*| \cos\left(\frac{\pi}{6} - \theta_\delta\right) \end{cases} \quad (37)$$

where $|u_s^*|$ can be obtained from (19).

The above three distances can represent the size embodiment of RVV in V_{opt1} , V_{opt2} , and $V_{0,7}$, respectively. Then, based on the above distance, the number of layers on these three voltage scales where the optimal VV is located can be expressed as

$$\begin{cases} c_1 = \text{ceil}\left(\frac{3\sqrt{3}d_1}{|u_{dc}|}\right) \\ c_2 = \text{ceil}\left(\frac{3\sqrt{3}d_2}{|u_{dc}|}\right) \\ c_3 = \text{ceil}\left(\frac{3\sqrt{3}d_3}{|u_{dc}|}\right) \end{cases} \quad (38)$$

where the “ceil” is the up-integer function and c_1 , c_2 and c_3 correspond to the layers numbers of the RVV on V_{opt1} , V_{opt2} , and $V_{0,7}$, respectively, and also reflect the proportionality of the three vectors.

As shown in Fig. 5, when the c_1 layer is 1, the c_2 layer is 2, and the c_3 layer is 3, then, the gray TS in the Fig. 5 is the optimal TS. When the optimal TS is selected, and the SCVV located in this TS is the optimal VV. This process transforms the selection of the optimal VV from the quantitative traversal of the cost function to a geometric topological relationship.

Based on the number of layers mentioned in (38), the action time of the SCVVs can be calculated in proportion to the individual basis vectors as

$$\begin{cases} t_{opt1} = \left(\frac{1}{3}c_2 - \frac{2}{9}\right)T_s \\ t_{opt2} = \left(\frac{1}{3}c_1 - \frac{2}{9}\right)T_s \end{cases} \quad \text{Case A}$$

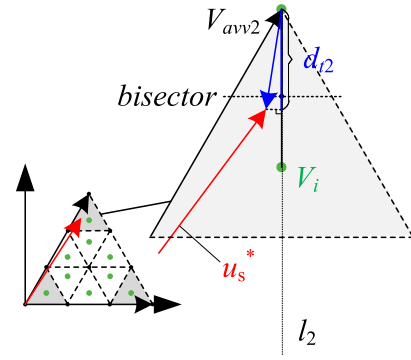


Fig. 6. Selection mode of the basic VVs.

$$\begin{cases} t_{opt1} = \left(\frac{1}{3}c_2 - \frac{1}{9}\right)T_s \\ t_{opt2} = \left(\frac{1}{3}c_1 - \frac{1}{9}\right)T_s \end{cases} \quad \text{Case B} \quad (39)$$

$$t_z = T_s - t_{opt1} - t_{opt2} \quad (40)$$

where t_{opt1} , t_{opt2} , and t_z are acting time intervals of the first operating VV, the second operating VV, and the zero VV, respectively. When $(c_1 + c_2 + c_3)$ is odd number, Case A is used and Case B is used when $(c_1 + c_2 + c_3)$ is even number.

From above, the optimal SCVV can be determined based on the operating VVs, presented in Table I and their corresponding action times, presented in (39) and (40).

The basic VVs within this sector, i.e., the two active VVs $V_{avv1,2}$ and zero VV $V_{0,7}$, are also considered to ensure the dc voltage utilization. In the TSs terminated by the basic VVs, it is necessary to compare SCVVs with the basic VVs. Taking the basic VV V_{avv2} as an example, the following selection method is designed based on the distances mentioned in (37).

As shown in Fig. 6, since the auxiliary line l_2 coincides with the straight line where $V_{avv2} - V_i$ is located, the projection of the difference VV between the RVV u_s^* and V_{avv2} in l_2 is defined as d_{t2} . Then, according to the geometric relationship, the distance between V_{avv2} and V_i is $2\sqrt{3}|u_{dc}|/27$. Obviously, when d_{t2} is less than half of the distance between V_{opt2} and V_i , i.e., $\sqrt{3}|u_{dc}|/27$, the V_{avv2} is closer to RVV and selected as the optimal VV. The other two basic VVs are selected by similar principles.

According to (37), the distances d_{t0} , d_{t1} , and d_{t2} , which can evaluate the three VVs $V_{opt0,7}$, V_{opt1} , and V_{opt2} are

$$\begin{cases} d_{t0} = d_3 \\ d_{t1} = \frac{\sqrt{3}}{3}|u_{dc}| - d_1 \\ d_{t2} = \frac{\sqrt{3}}{3}|u_{dc}| - d_2. \end{cases} \quad (41)$$

Noteworthy, the selection of the basic VVs needs to be operated before the selection of the center VVs of the TSs, i.e., if any of the distances mentioned in (41) satisfy less than $\sqrt{3}|u_{dc}|/27$, then the corresponding basic VV is selected directly without the later process about selecting the center VVs of the triangular subarea. This not only ensures the logical correctness of the selection of the optimal VV, but also further enhances the simplicity of the algorithm.

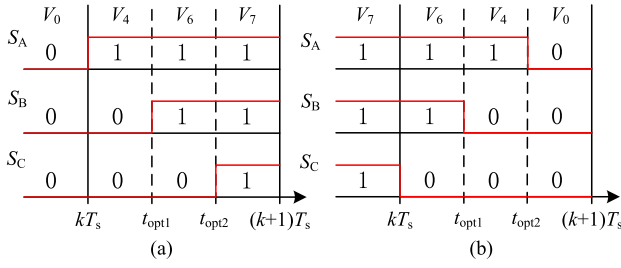


Fig. 7. Proposed switching pattern optimization method for the SCVV in Sector I (a) $S_{A,B,C} = 000$. (b) $S_{A,B,C} = 111$.

B. Switching Patterns Optimization

To reduce the switching frequency of the SCVV, a switching pattern optimization method is designed in this article. Noteworthy, this switching pattern optimization method only optimizes the SCVV since the basic VV do not need to focus on the switching sequence.

The two active VVs are operated first and the zero VV is operated latter. To take the switching state of the previous moment into account, the output switching pattern of this control period needs to be constructed according to the zero VV of the previous control period. The first operating VV of this control period selects the active VV that only switches once compared with the zero VV of the previous instant. Meanwhile, the zero VV of this control period depends on the second operating vector, and the one with fewer switching actions is selected. Because the two active VVs can always be divided into two cases: case I: $S_A + S_B + S_C = 1$ and case II: $S_A + S_B + S_C = 2$. So, the first active VV can be determined by the following logic:

$$V_{avv1} = \begin{cases} \text{case I,} & \text{if previous Zero VV} = V_0 \\ \text{case II,} & \text{if previous Zero VV} = V_7 \end{cases} \quad (42)$$

where V_{avv1} is the first active VV. Then, the second active VV should be another one and the zero VV in this control period can be determined by

$$\text{Zero VV} = \begin{cases} V_0, & \text{case I} \\ V_7, & \text{case II.} \end{cases} \quad (43)$$

Taking the RVV in Sector I as an example, as shown in Fig. 7(a), if the previous control period is V_0 (0, 0, 0), AVV1 in this control period select V_4 (1, 0, 0), and AVV2 is V_6 (1, 1, 0). At the same time, the final vector of this control period, the ZVV, should select the minimum switching vector according to AVV2 as V_7 (1, 1, 1) instead of V_0 (0, 0, 0). Another case is also shown in Fig. 7(b).

Above all, the VV selection process can be briefly described as follows: after obtaining the RVV information through the model-free approach, first, the active vectors V_{avv1} and V_{avv2} are determined according to the reference angle, and then the projection distances $d_{1,2,3}$ to the three auxiliary lines $l_{1,2,3}$ are determined through (37). Second, the basic optimal VV is selected according to the distances $d_{1,2,3}$ representing the individual basic VV, and if any of them satisfy the conditions, the switching sequence is directly output. If the conditions are not satisfied, the number of layers $c_{1,2,3}$ where the optimal

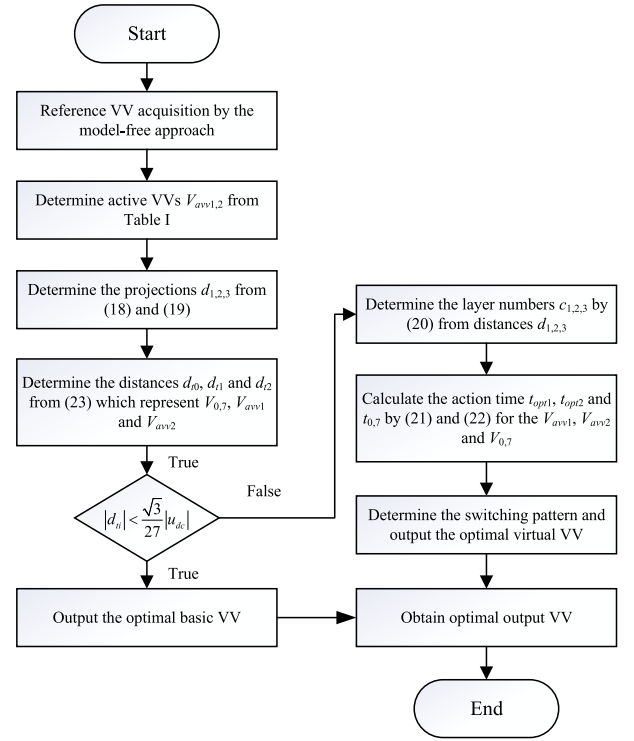


Fig. 8. Flow diagram of the proposed VV selection principle.

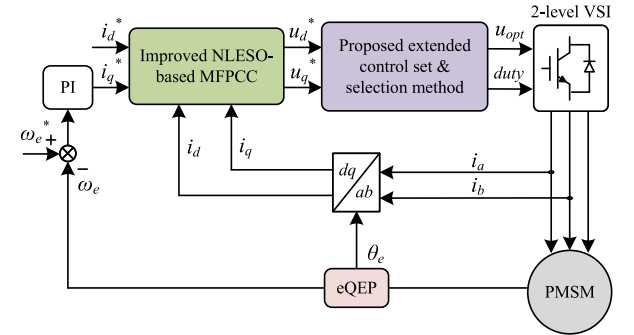


Fig. 9. Schematic of the proposed MFPC method.

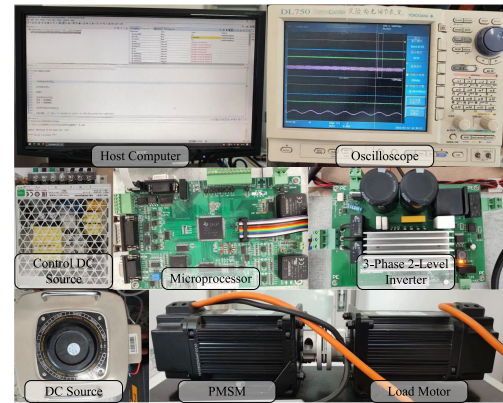


Fig. 10. Experiment platform of the SPMSM drive system.

voltages are located is determined according to $d_{1,2,3}$, and the duty cycles of the corresponding vectors t_{opt1}, t_{opt2} , and t_{opt3} are determined according to the number of layers. Finally, the

TABLE II
NOMINAL PARAMETERS OF THE SPMSM
IN THE PLATFORM

Parameters	Value
Pole pairs N_p	4
Rotational inertia J	$2.0 \times 10^{-3} \text{ kg} \cdot \text{m}^2$
Stator resistance R_s	0.90Ω
d/q axis inductance L_s	5.5 mH
Permanent magnet flux ψ_f	0.11 Wb
Rated power	0.75 kW
Rated speed	$3.0 \times 10^3 \text{ r/min}$
Rated torque T_e	2.0 Nm

output sequence of the optimal virtual voltages is determined through optimization. After the above selection process, this proposed optimal VV selection method can directly determine the optimal VV among the 62 candidate VVs in the newly designed extended control set, which effectively simplifies the optimal VV selection process and reduces the algorithm complexity while ensuring high output voltage accuracy. Fig. 8 shows the flow diagram of the proposed extended control set and its selection method, and Fig. 9 shows the schematic of the proposed MFPC method.

V. EXPERIMENTAL VERIFICATION

The feasibility of the proposed MFPC method was investigated experimentally on a 0.75 kW SPMSM platform. The platform setting is shown in Fig. 10. The specifications of the SPMSM parameters are provided in Table II. The TMS320F28335 microprocessor from Texas Instruments (TIs) performs the control strategies. The sampling frequency of the methods below is 10k, and the current limitation value is 6 A. The experimental results are collected by the oscilloscope and transferred to the PC for better display and further analysis. For simplicity, “Method I” stands for the conventional MPCC method, the proposed NLESO-based model-free method with the DSVM method proposed in [30] is referred to “Method II”, and “Proposed Method” stands for the MFPC method in this article. As discussed in Section III-B, for maintaining chatter avoidance and strong robustness, the parameters of observer are set the same as $\alpha = 1/L_s$, $\varepsilon = 0.5$, and $\delta = 0.01$. According to [28], the parameter ω_0 of the conventional NLESO and the proposed one are set to the same 1250 rad/s. For the methods presented in this section, the control principle of $i_d = 0$ was used to satisfy the maximum torque per ampere principle. In this article, the standard deviation is used to measure the fluctuation of dc variables.

A. Steady-State Performance With Nominal Parameters

This section conducts relevant experiments on the steady-state performance of Method I, Method II, and the Proposed Method to verify the effectiveness of the proposed extended control set in improving the steady-state performance of MFPC. The parameters in the above methods are set to nominal ones, as shown in Table II. The steady-state performance waveforms at 300 r/min speeds with 0.5 N·m

load, 1500 r/min speeds with 1 N·m load, and 3000 r/min with 2 N·m load are shown in Figs. 11–13, respectively. As shown in Fig. 11, under 300 r/min speed and 0.5 N·m load conditions, steady-state control performance is significantly improved by using the extended control set methods, Method II, and Proposed Method, compared to the conventional FCS-MPC method, Method I, which uses the basic VVs as the candidate VVs. Moreover, the extended control set proposed in this article has better steady-state control performance than the conventional DSVM-based method, Method II. In particular, the Proposed method has an 11.30% lower q -axis current fluctuation, a 15.35% lower d -axis fluctuation, and a 14.93% lower phase current THD than Method II. Similarly, this conclusion can be obtained from the experimental results at 1500 r/min speeds with 1 N·m load and 3000 r/min with 2 N·m load working conditions. Whether d -axis, q -axis current, or phase current THD, the proposed extended control set method has different degrees of enhancement in steady-state performance compared to the conventional DSVM. The phase current THD exhibited by the proposed extended control set is reduced by 9.79% compared to the conventional DSVM at 1500 r/min speeds with 1 N·m load, while it is reduced by 6.53% at 3000 r/min with 2 N·m load. In fact, consistent with the previous analysis, the reason for the better steady-state performance of the proposed method is that the VVs in the proposed extended control set enable the output voltage to have a higher accuracy.

B. Computational Burden Comparison

As analyzed in Section IV-B and experimental results in Section V-A, extended more additional candidate VVs to the control set can improve the steady-state control performance by enhancing the output voltage accuracy, however, it inevitably increases the difficulty of selecting the optimal VV. To demonstrate the effectiveness of the proposed geometry-based VV selection method, the computational burden performance of Method I, Method II, and the Proposed Method are compared and the results are presented in Fig. 14. To ensure a fair comparison, the execution time of each method is the average of several experiments under different working conditions. As shown in Fig. 14, Method I has less execution time than the other two methods because only basic VVs are evaluated. However, the very limited quantity of candidate VVs will result in large current fluctuations and poor steady-state performance, verified in Section V-A. Meanwhile, the implementation time of Proposed Method is 23.40% less than the Method II. This indicates that the reduced computational burden ensures the simplicity of the algorithm with improved steady-state performance.

C. Switching Frequency Comparison

The increase in switching frequency leads to additional switching losses which will decrease the efficiency. In this section, we evaluate the proposed method switching frequency. To better evaluate the switching frequency, this article

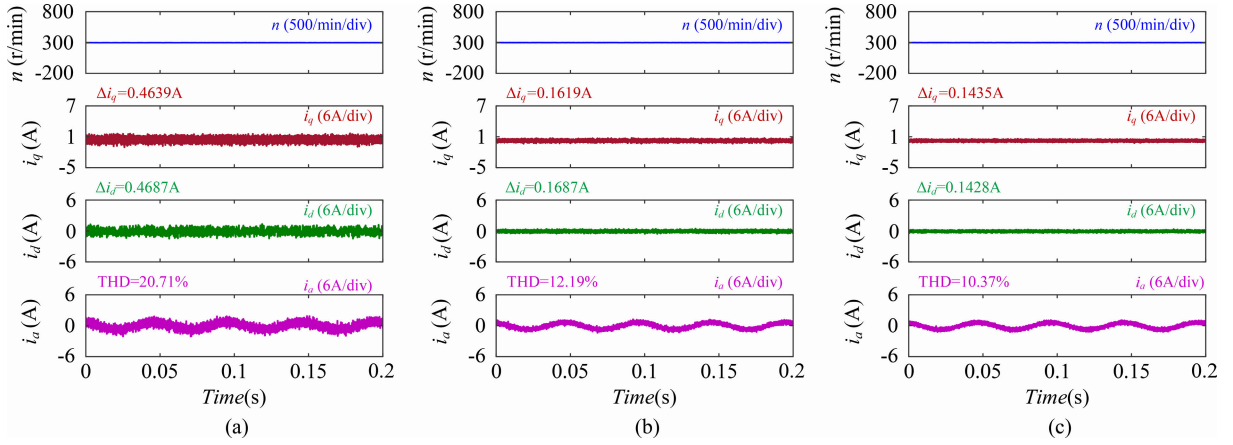


Fig. 11. Experimental results of speed, d - and q -axis current and phase current under different methods at 300 r/min and 0.5 N-m. (a) Method I. (b) Method II. (c) Proposed Method.

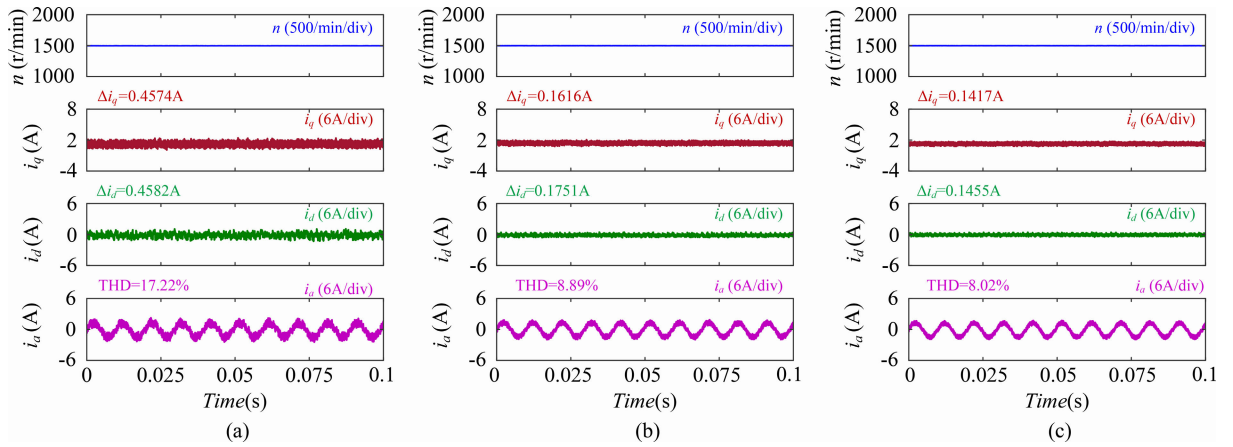


Fig. 12. Experimental results of speed, d - and q -axis current and phase current under different methods at 1500 r/min and 1 N-m. (a) Method I. (b) Method II. (c) Proposed Method.

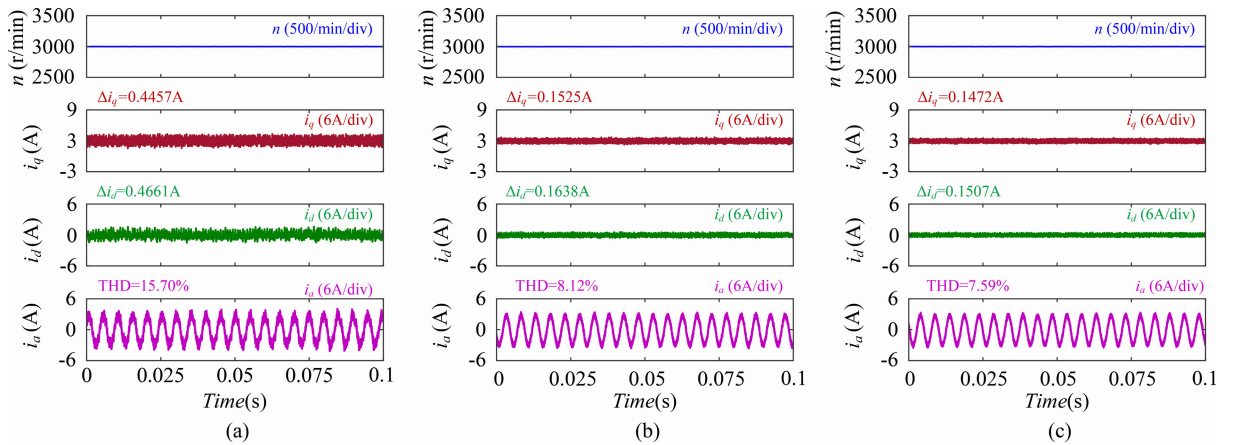


Fig. 13. Experimental results of speed, d - and q -axis current and phase current under different methods at 3000 r/min and 2 N-m. (a) Method I. (b) Method II. (c) Proposed Method.

introduces the average switching frequency (ASF), defined as

$$f_{ASF} = N_{total}/6T_m \quad (44)$$

where f_{ASF} is the ASF and N_{total} is the total switching actions within the time range, i.e., T_m .

Fig. 15 compares the ASF between the proposed method with and switching pattern optimization method under

different speeds and load torque conditions. As shown in Fig. 15, after using the switching pattern optimization method in Section IV-B, there is a reduction in the switching frequency of the proposed method compared to the switching pattern without optimized. Correspondingly, the efficiency of the motor under different operating conditions after using the switching pattern optimization method is shown in Fig. 16.

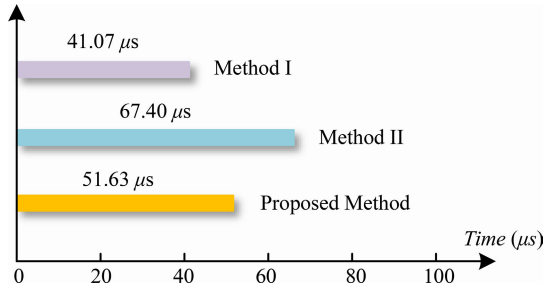


Fig. 14. Execution time of Method I, Method II, and proposed method.

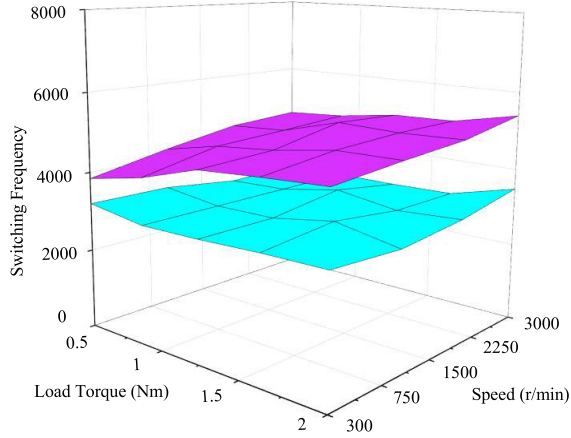


Fig. 15. ASF comparison between the proposed method with and without switching pattern optimization.

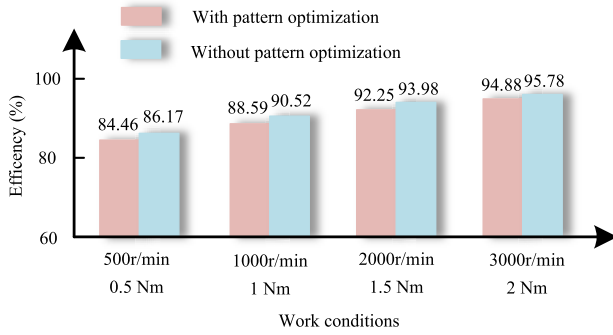


Fig. 16. Efficiency comparison among the proposed method with and without switching pattern optimization.

The results show that the switching losses of the motor are decreased after using the switching sequence optimization method as the reduction of the switching frequency.

D. Robustness Performance Comparison Under the Mismatch Parameters

To evaluate the robust performance of the proposed method, in this section, the performance of conventional nonlinear ESO and observer-based MFPCC under parameter mismatch conditions is verified. The modulation method proposed in this article was used in this comparison experiment.

Estimation performance experiments in the parameter mismatch situation are executed to verify the performance of the proposed smooth nonlinear ESO. In this section, the proposed

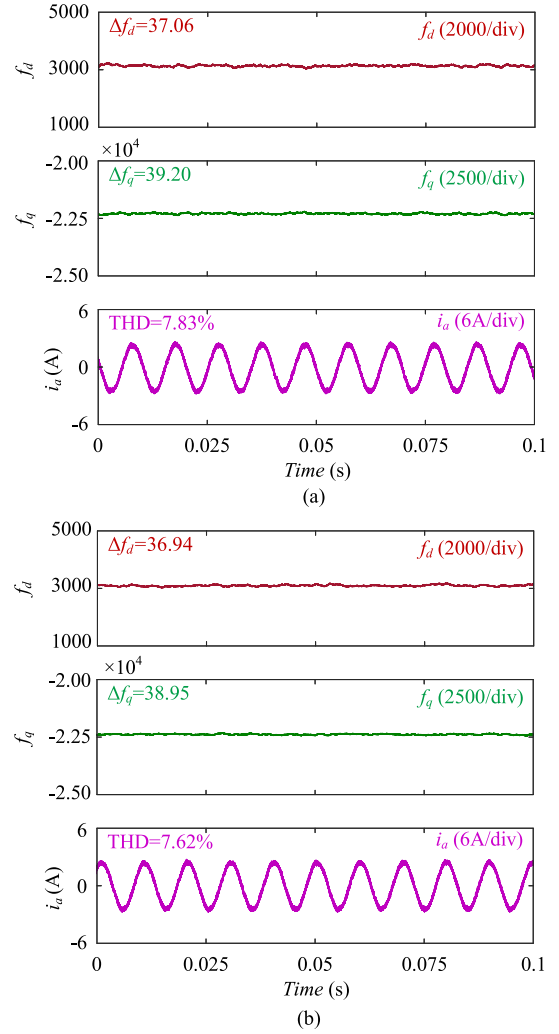
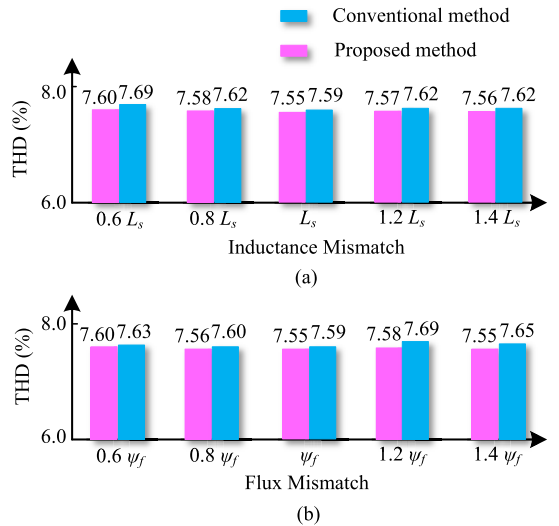
Fig. 17. Experimental results for parameter mismatch of $0.6 L_s$ and $1.4 \psi_f$ at the speed of 1500 r/min and the load of 2 N-m. (a) Conventional NLESO method. (b) Proposed method.

Fig. 18. Experimental results of the proposed current control performance under different parameter mismatches (a) inductance mismatches (b) flux mismatches.

method of extended control set is used uniformly to control the effects of other variables. Then, the conventional nonlinear

TABLE III
PERFORMANCE OF THE PROPOSED METHOD UNDER
THE DIFFERENT PARAMETER ε

Parameters	$\varepsilon=0.1, \delta=0.01$	$\varepsilon=0.5, \delta=0.01$	$\varepsilon=0.75, \delta=0.01$
Δf_d	19.06	19.05	19.07
Δf_q	21.31	20.34	20.36
THD (%)	7.70	7.55	7.61

TABLE IV
PERFORMANCE OF THE PROPOSED METHOD UNDER
THE DIFFERENT PARAMETER δ

Parameters	$\varepsilon=0.5, \delta=0.001$	$\varepsilon=0.5, \delta=0.01$	$\varepsilon=0.5, \delta=0.1$
Δf_d	19.05	19.05	19.09
Δf_q	20.37	20.34	23.41
THD (%)	7.62	7.55	7.83

ESO is compared with the smooth nonlinear function-based NLESO proposed in this article. At the same time, the working conditions were set to the same at a speed of 1500 r/min and a load of 2 N·m.

The experimental results under the parameter mismatch at $0.6 L_s$ and $1.4 \psi_f$ for the nonlinear observer are shown in Fig. 17(a), and the smooth function results are shown in Fig. 17(b). From the experimental results, the fluctuation of the observed lump disturbance of the observer using the smooth nonlinear function fald in this article is smaller than that of the observer using the fal nonlinear function. In addition, the current THD value corresponding to the control method with the fal nonlinear function observer is also smaller than that using the fal nonlinear function observer. To further verify the effect of the smooth nonlinear function-based ESO on the control performance, the current control performance of the two observers under different parameter mismatch conditions is shown in Fig. 18. From the experimental results, the control method with the smooth nonlinear function fald have the better estimation and robustness performance.

E. Comparative Current Tracking Response Performance

This section compares the current tracking performance with sudden load from 0.5 to 2 N·m to further validate the proposed observer-based MFPC method's performance advantages compared to the conventional nonlinear ESO-based MFPC method. In this comparison experiment, the speed was set to 1500 r/min and the load was increased from 0.5 to 2 N·m starting at 0.05 s. The mismatch condition is set to 0.6 times L_s and 1.4 times the ψ_f . The waveforms of the two methods mentioned above are presented in Fig. 19. From Fig. 19, the dynamic rise time of the proposed method is relatively shorter compared with the conventional method, which demonstrates superior current tracking performance due to the high estimation performance of the proposed NLESO.

F. Parameter Sensitivity Analysis of Proposed NLESO Parameters

This section implements experiments at different parameter settings to evaluate the effect of each parameter. To prevent

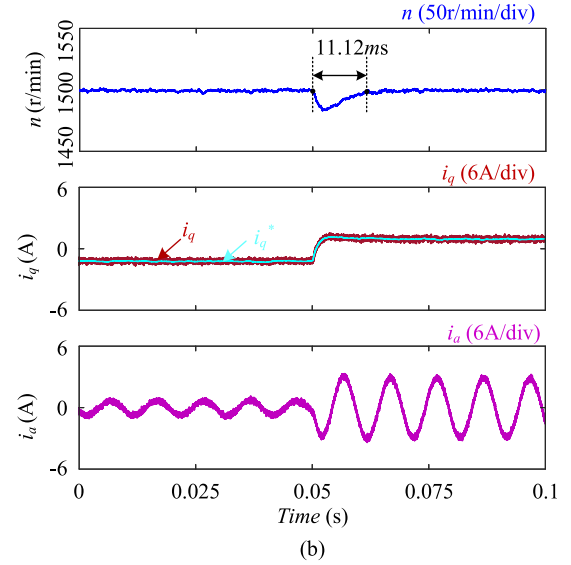
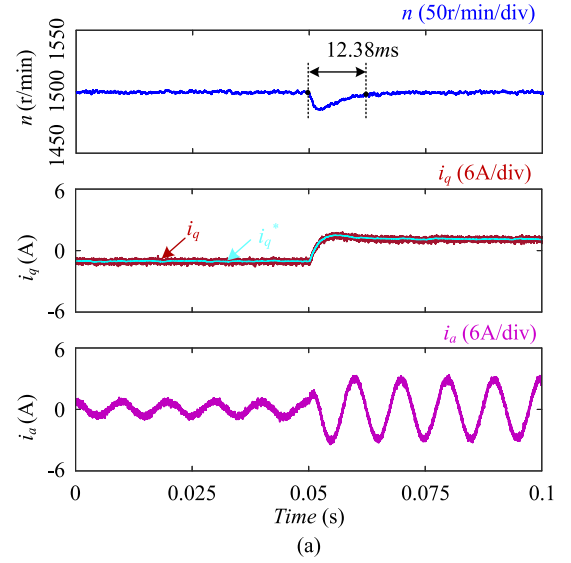


Fig. 19. Experimental results of the sudden load torque from 0.5 to 2 N·m at speed 1500 r/min with $0.6 L_s$ and $1.4 \psi_f$ parameter mismatch. (a) Conventional NLESO method. (b) Proposed Method.

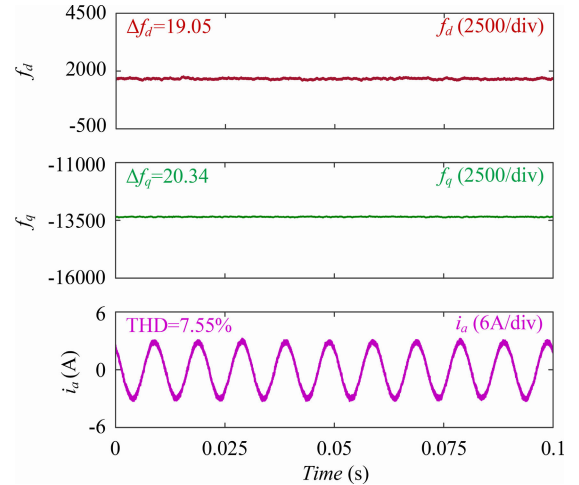


Fig. 20. Estimation performance of the proposed smoothness-based ESO at $\varepsilon = 0.5, \delta = 0.01$ parameter setting.

the influence of other factors, the experimental results in this section are under nominal parameters and run at a

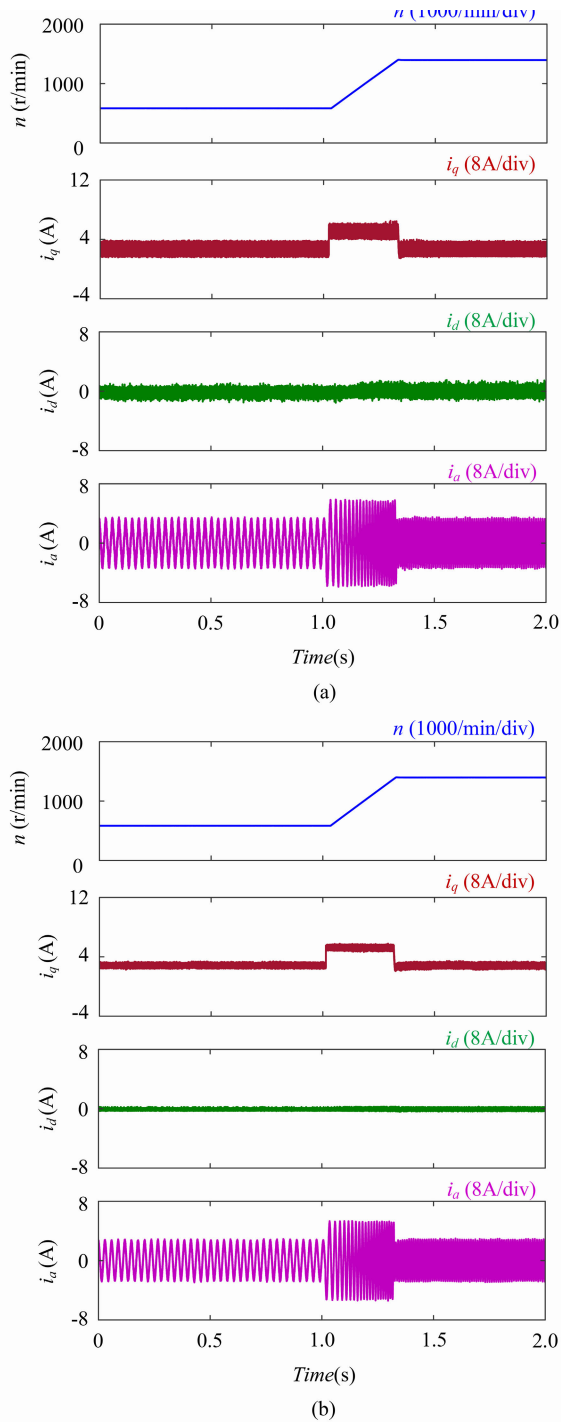


Fig. 21. Experimental results of step speed up from 500 to 1500 r/min at 2 N·m torque. (a) Conventional MPCC. (b) Proposed Method.

speed of 1500 r/min and a load of 2 N·m for the working conditions.

The experimental results for the parameter settings of $\alpha = 0.5$ and $\delta = 0.01$ are displayed in Fig. 20. The experimental results related to different α at $\delta = 0.01$ are presented in Table III to facilitate the comparison of the effects of different parameters. According to the data in Table III, choosing $\alpha = 0.5$ has better estimation performance under this experimental condition compared to other parameter settings. Similarly, the

experimental results for different δ parameters are presented in Table IV, which show that either too large or too small a δ value will deteriorate the estimation performance of the proposed NLESO and, consequently, weaken the current control performance of the control system.

G. Speed Tracking Response Performance

Step-speed response comparison experiments are conducted in this section to verify the proposed method's speed control performance in this article. Remarkably, nominal parameters were used in the experiments to control for variable effects and better compare speed performance. The experiment with the step speed from 500 to 1500 r/min in 2 N·m load was performed, and the results were shown in Fig. 21, where Fig. 21(a) shows the speed performance of the conventional MPC method and Fig. 21(b) shows the proposed method in this article. Then, according to the above experimental results, the proposed method can achieve better state-state performance under similar dynamic control performance.

VI. CONCLUSION

This article proposes an MFPC method using a high-smoothness nonlinear ESO based on a newly designed extended control set. By introducing the designed new nonlinear function with the smooth polynomial series into the NLESO framework, the estimation performance of the unknown part F is improved. Then, the observation performance is enhanced. In addition, based on the DSVM theory, the steady-state performance of MFPC is improved by the newly designed extended control set, which contains more VVs. Moreover, based on the geometrical characteristics of SCVVs, the fast-layering VV selection method simplifies the difficulty of traversing the new extended control set and reduces the computational burden of the algorithm execution. In addition, the switching patterns of virtual VVs are optimized to achieve less switching frequency and lower switching loss.

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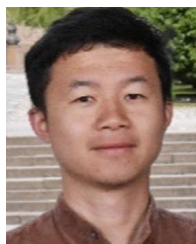
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